



CSSE Faculty and Alumni Publish on Tennis Performance and Injury Risk

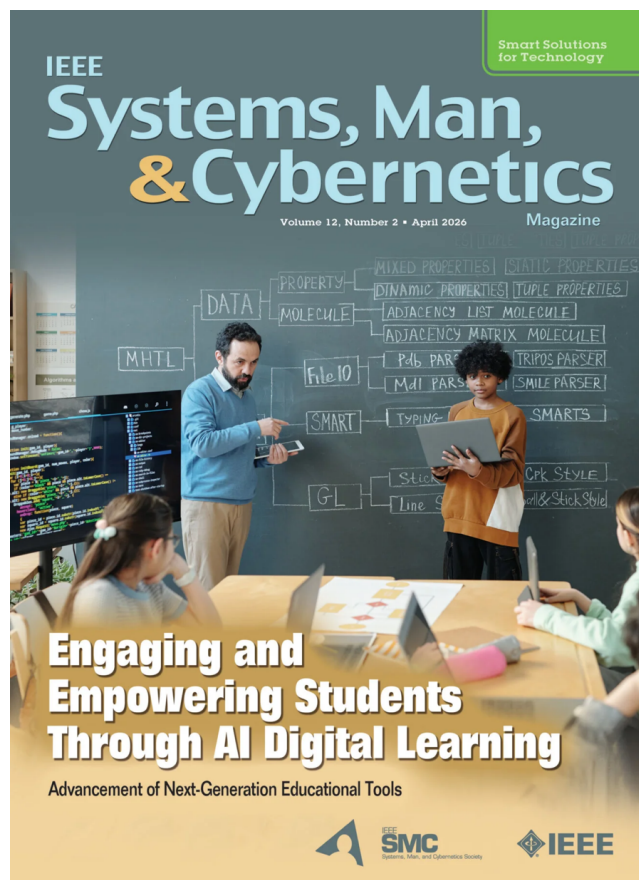
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Department of Computer Science and Software Engineering (CSSE) faculty Weihao Qu, Ph.D., Ling Zheng, Ph.D., and Jiacun Wang, Ph.D., along with recent alumni Dongyang Wang '25M, Francisco E. Alvarez '25M, and **Shobharani Polasa** '24M, published research in [Institute of Electrical and Electronics Engineers \(IEEE\) Systems, Man, and Cybernetics magazine](https://ieeexplore.ieee.org/xpl/tocresult.jsp?isnumber=9606875) (<https://ieeexplore.ieee.org/xpl/tocresult.jsp?isnumber=9606875>).

Their article, “[Multimodal Injury Risk and Performance Prediction in Tennis Using Weighted Ensemble Learning](https://ieeexplore.ieee.org/abstract/document/11513671) (<https://ieeexplore.ieee.org/abstract/document/11513671>),” explores how machine learning can help predict tennis players’ performance and injury risk by integrating data from multiple sources into a single system.

“Our goal is to develop a data-driven AI framework that provides a comprehensive Athlete Readiness Score and personalized injury risk assessment for tennis players ... We hope this work can help coaches and players identify potential injury risks early and make better training and recovery plans,” Dongyang Wang said.

The authors introduce Predictive Athlete Readiness for Tennis (PART), a multimodal weighted ensemble learning framework. PART combines data from wearable devices, surveys, physical tests, match information, and video analysis to assess player readiness, performance potential, and injury risk more accurately.



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